Fiscal Year:	FY 2016	**Task Last Updated:**	FY 08/31/2016
PI Name:	Sabanayagam, Chandran Ph.D.		
Project Title:	Epigenetic and Protein Expression Pattern Profiling of Caenorhabditis elegans Exposed to Time-Varying Gravitational Fields: a Multi-Generational Study		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Developmental biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Animal Biology: Invertebrate		
Space Biology Cross-Element Discipline:	(1) Reproductive Biology (2) Developmental Biology (3) Musculoskeletal Biology (4) Neurobiology		
Space Biology Special Category:	None		
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Zip Code:	19711-5449	**Congressional District:**	1
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2012 Space Biology NNH12ZTT001N
Start Date:	09/01/2013	**End Date:**	08/31/2017
No. of Post Docs:	1	**No. of PhD Degrees:**	0
No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	1
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA ARC
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Flight Program:			
Flight Assignment:	NOTE: Extended to 8/31/2017 per NSSC information (Ed., 9/1/16)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NNX13AM08G		
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Task Description:	We hypothesize that long term temporal changes in the gravity vector will result in animal adaptation via epigenetic mechanisms that are passed onto subsequent generations. We further hypothesize that the gravity-influenced epigenetic profile can be restored to the consensus wild-type (i.e., Earth born) upon introduction to stable 1xg conditions. We propose to utilize C. elegans as a model organism to address this hypothesis by performing the following experiments: 1) Determine C. elegans behavioral responses (i.e., changes in migration dynamics, animal body posture, metabolism, etc.) due to temporal changes in the gravity vector. The gravity vector will be controlled by placing the animals inside a rotating CubeSat microscope developed by our group. The frequency, direction, and duration of rotation will be varied to elicit the most dramatic behavioral responses. Habituation time will be evaluated, and also changes in lifespan. 2) Identify gravity responsive genes by utilizing the vast collection of fluorescent strains of C. elegans available to researchers. We will monitor gene expression by fluorescence microscopy--from the embryonic stage through adulthood--during changes in the gravity vector. Genes of interest include those associated with muscle development, cell apoptosis, and neuronal development. 3) Profile the epigenetic marks on C. elegans exposed to long periods of continuous gravity vector changes that range from 1/4 of an animal's lifespan to three generations. We will examine the epigenetic consequence of restoring constant 1xg conditions over successive generations. The three Specific Aims can be conducted with the same animal strains, and thus allows the opportunity to correlate the findings across different experiments.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Understanding gravity's affect on biological systems is not well studied because this parameter is difficult to manipulate during experiments. The International Space Station houses the newest U.S. national laboratory, which can be utilized to understand the influence of microgravity on molecular, cellular, and animal model systems. Determining changes in epigenetic profiles during multi-generations exposed to different environments is important to know to prepare for future manned spaceflight beyond Earth. Additionally, temporary alteration of epigenomes is relevant to human health, and similar adaptation mechanisms may be displayed by almost any change in the environment or diet. In order to expedite molecular and cellular research in microgravity, we developed an innovative clinorotation microscope that enables us to observe living organisms, in real-time, during microgravity. Because of the extended clinorotation times (4 days to 3 months), we also developed a novel automated "worm chip" that permits hands-free feeding and maintenance of the animals, without interrupting clinorotation.
Task Progress:	Year 3 Goals and Objectives: * Perform altered gravity experiments on animals over multi-generations; * Examine the phenotype of animals exposed to altered gravity using strains that express fluorescent proteins to genes identified by bioinformatics. Year 3 Outcomes: We have made significant progress with our genetic and microscopy experimental procedures, and have a solid bioinformatics pipeline in place. Several highlights are presented in this interim report. In particular, we are the first to validate the use of clinorotation to simulate microgravity by comparing our genetic data with space-flown animals. Furthermore, our bioinformatics pipeline has revealed several pathways that are altered when exposed to simulated microgravity, including longevity and dorsal-ventral axis determination. Our group has recently coined the term, “gravitome,” to describe the collection of genes that are responsive to changes in the gravitational vector. * The genetic effects of simulated microgravity persist for ~3 generations. We have completed a multi-generational analysis of the transcriptome of C. elegans exposed to simulated microgravity. P0 animals acclimated to liquid (> 1 month) were exposed to simulated microgravity for 10 days. Immediately following the simulation, RNA was harvested from the F1 animals. RNA samples were also harvested from animals over three generations after microgravity exposure (F2, F3, and F4). * Core genes identified for C. elegans. We have conducted many adaptation experiments in this project, including culturing agar-grown animals in liquid, changing the animals’ diet from bacterial source to a synthetic medium (CeHR), and exposing the animals to simulated microgravity. These varied environmental conditions have allowed us to determine a set of core genes, defined as having a relatively constant expression profile (-1 < log2[expression ratio] < 1) over all conditions examined. We hypothesize that these core genes are necessary for animal survival, and altering their expression levels should have a profound consequence on their health. * Longevity: Sphingolipid metabolic pathways are altered in simulated microgravity. Previous spaceflight experiments suggest that key longevity-regulating transcription factors DAF-16/FOXO and SKN-1 are responsive to gravity. Our bioinformatics analysis also show differential regulation of longevity pathways, and moreover we have identified sphingolipids as the putative substrate that is targeted by these genes. Our analysis has not only produced a list of differentially expressed genes, but also a list of altered pathways and associated pathway products. We now hypothesize that decreased concentrations of sphingolipids will lead to longer lifespan in our simulated microgravity experiments. We are now in the process of directly measuring the animals’ lifespan in simulated microgravity to test this hypothesis. * Dorsal-Ventral determination pathway is altered in simulated microgravity. Another interesting pathway that is altered during simulated microgravity is body axis determination. Although it is not clear how this pathway affects the phenotype (animals develop normally), it may be the case that the key checkpoint is during embryogenesis, because Grk is expressed in both oocytes and dorsal follicle cells. We will test animal fecundity during simulated microgravity to determine if this condition produces less healthy embryos. Intuitively, it seems reasonable to expect that gravity plays a major role in body axis determination and body symmetry. * Awakening of dormant genes through adaptation. Our study provided transcript evidence for a number of protein coding genes (CDSes) that have not been experimentally verified. Considerable effort has been devoted to annotating the genes in C. elegans, but the standard WormBase annotation still contains thousands, about 8%, of the of CDSes for which there are no direct experimental evidence. Many of these unsupported gene models or undetected genes in the C. elegans genome have not been identified by computational let alone experimental means because they are either poorly expressed, have weak statistical signals, or are less conserved across species. We entertain another possibility that these undetected transcripts simply need a trigger to activate their expression, and this trigger may be elicited when animals adapt to certain environment stimuli. Collectively, our adaptation experiments have verified an overwhelmingly 22% (661) of all the previously unconfirmed genes in C. elegans, a significant contribution to the field!

	* Simulated microgravity leads to loss of muscle mass. Although loss of muscle mass is well documented in spaceflight research, we used green fluorescent protein (GFP) markers in muscle-related proteins as a semi-quantitative method to identify similar muscle mass changes during clinorotation. Indeed, clinorotated animals showed decrease in muscle genes compared to ground control animals grown in liquid. In this case, clinorotation was performed for three weeks. paqr-2 is an ortholog of human ADIPOR1 (adiponectin receptor 1) and ADIPOR2 (adiponectin receptor 2) and is involved in positive regulation of fatty acid oxidation, reproduction, response to cold, and tail spike morphogenesis. paqr-2 is expressed in the nervous system, seam cell, gonadal sheath cell, and the head muscle, and is localized to the plasma membrane. Our previous bioinformatics analysis have revealed differential expression in vulva-related genes, and GFP microscopy shows an abnormal and protruding vulva. F25H2.1 is an ortholog of human RRNAD1 (ribosomal RNA adenine dimethylase domain containing 1) and is localized to the striated muscle dense body. Ground controls shows gene expression primarily in the head region and punctuated marks along the body wall. Clinorotated animals show pronounced expression in the head region, and loss of punctuated marks along the body. Animals removed from clinorotation and exposed to ground conditions for one generation (reversion) display GFP expression profiles similar to ground controls. myo-3 encodes MHC A, the minor isoform of MHC (myosin heavy chain) that is essential for thick filament formation, and for viability, movement, and embryonic elongation. The gene is expressed in body muscle, the somatic sheath cell covering the hermaphrodite gonad, and also expressed in enteric vulva muscles of the hermaphrodite and the diagonal muscles of the male tail. During clinorotation, we noted a qualitative difference in the locomotion of animals: after about three days of clinorotation, the animals appeared to be lethargic, with little movement compared to ground control animals that thrash and swim. Exposure to clinorotation decreased the amount of MHC A (reduced fluorescence and shorter striated marks); however, reverting the animals back to ground conditions restored the striated muscle phenotype. Student Awards: The Department of Biological Sciences at the University of Delaware awarded Andrew Moore a travel award to present his space biology research at the Biophysical Society Annual Meeting. Mr. Moore is a Masters student in Biology. The Education Committee of the Biophysical Society awarded Irem Celen a travel award to attend the annual meeting to present her space biology research. Ms. Celen is a Ph.D. student in Bioinformatics. Irem Celen also received a competitive Graduate University Fellow Award for her dissertation research on space biology.
Bibliography Type:	Description: (Last Updated: 08/17/2023)
Abstracts for Journals and Proceedings	Celen I, Doh JH, Sabanayagam CR. "Multi-scale spatio-temporal dynamics of histone modifications." Presented at the 2016 Biophysical Society Annual Meeting (invited talk), Los Angeles, California, February 27-March 2, 2016. Biophysical Journal. 2016 Feb 15;110(3, Suppl 1):350a. http://dx.doi.org/10.1016/j.bpj.2015.11.1885 , Feb-2016
Abstracts for Journals and Proceedings	Moore A, Celen I, Moore MT, Sabanayagam CR. "Wormpharm: A microfluidic platform for pharmacogenetic studies on C. elegans." 2016 Biophysical Society Annual Meeting, Los Angeles, California, February 27-March 2, 2016. Biophysical Journal. 2016 Feb 15;110(3, Suppl 1):504a-505a. http://dx.doi.org/10.1016/j.bpj.2015.11.2695 , Feb-2016
Articles in Peer-reviewed Journals	Doh JH, Moore AB, Celen I, Moore MT, Sabanayagam CR. "ChIP and Chips: Introducing the WormPharm for correlative studies employing pharmacology and genome-wide analyses in C. elegans." Journal of Biological Methods. 2016;3(2):e44. Published online 2016-06-10. http://dx.doi.org/10.14440/jbm.2016.109 , Jun-2016